

Linear equations with one unknown

One standard form, four steps, and the three things that can happen at the end.

LESSON 123–124

2 × 40 MIN

MATEMATIKA 7, §55

S8 2.5

LESSON CLOCK

10'

25'

25'

20'

5'

The standard form	10 min
The four steps	25 min
Brackets and fractions	25 min
The three endings	20 min
Homework	5 min

LEARNING OBJECTIVES

- Recognise a linear equation and write it as $ax = b$.
- Solve by collecting the unknowns on one side and the numbers on the other.
- Solve equations containing brackets and fractions.
- Interpret the cases $a = 0, b \neq 0$ and $a = 0, b = 0$.

TEXTBOOK

National	pp. 363–370
Cambridge	Stage 8, pp. 31–36
Workbook	Exercise 2.5

01

Explanation

The standard form

An equation is **linear** in x when, after simplification, it can be written as

$ax = b$

with no x^2 , no x in a denominator and no x under a root.

EQUATION	LINEAR?
$3x - 5 = 7$	yes
$2(x + 1) = 5x - 4$	yes — after expanding
$x^2 = 9$	no
$\frac{1}{x} = 3$	no — the unknown is in the denominator

WHY THE STANDARD FORM MATTERS

Once an equation reads $ax = b$ the answer is $x = \frac{b}{a}$ and nothing is left to decide. Every step of the solution is aimed at reaching that one line.

The four steps

STEP	WHAT TO DO	EXAMPLE: $3x - 5 = x + 7$
1	expand any brackets	none here
2	unknowns to the left, numbers to the right	$3x - x = 7 + 5$
3	collect like terms	$2x = 12$
4	divide by the coefficient	$x = 6$

A TERM THAT CROSSES THE EQUALS SIGN CHANGES ITS SIGN

-5 on the left becomes $+5$ on the right. Moving a term without changing its sign is the commonest error in the whole of equation-solving, and the check catches it every time.

Brackets and fractions

Expand first; clear denominators by multiplying **every** term by the lowest common denominator.

EQUATION	FIRST MOVE	THEN	ROOT
$2(x + 3) = 5x - 6$	expand	$2x + 6 = 5x - 6$	$x = 4$
$\frac{x}{2} + \frac{x}{3} = 5$	multiply by 6	$3x + 2x = 30$	$x = 6$
$\frac{x-1}{4} = \frac{x+1}{6}$	multiply by 12	$3(x-1) = 2(x+1)$	$x = 5$

EVERY TERM IS MULTIPLIED, THE WHOLE-NUMBER ONES TOO

In $\frac{x}{2} + \frac{x}{3} = 5$ the 5 becomes 30. Multiplying only the fractions is the standard slip and gives a wrong root that the check exposes at once.

The three endings

FINAL LINE	MEANING	ANSWER
$2x = 12$	$a \neq 0$	$x = 6$
$0 \cdot x = 5$	$a = 0, b \neq 0$	no roots
$0 \cdot x = 0$	$a = 0, b = 0$	every number is a root

And the check, always into the original equation:

EQUATION	ROOT	LEFT	RIGHT
$3x - 5 = x + 7$	$x = 6$	13	13
$2(x + 3) = 5x - 6$	$x = 4$	14	14

02

Worked examples

EXAMPLE 1

Solve $3x - 5 = x + 7$.

1

$3x - x = 7 + 5$
Both terms change sign as they cross.

2

$2x = 12$

3

$x = 6$

4

Check: $3 \cdot 6 - 5 = 13$ and $6 + 7 = 13$ ✓

ANSWER

$x = 6$

EXAMPLE 2 Solve $\frac{x}{2} + \frac{x}{3} = 5$.

- ① Multiply every term by 6.
Including the 5.

② $3x + 2x = 30$

③ $5x = 30$

④ $x = 6$

ANSWER

$x = 6$

EXAMPLE 3 Solve $2(x + 3) = 2x + 5$.

- ① $2x + 6 = 2x + 5$
Expand.

② $2x - 2x = 5 - 6$

③ $0 \cdot x = -1$

- ④ No value of x can satisfy this.

ANSWER

No roots

03 Interactive model

Insist on the check for one full lesson and mark it as a separate step; the number of sign errors handed in falls immediately.

One step at a time

In $3x - 5 = x + 7$, moving x to the left gives:

Moving -5 to the right gives:

So the equation becomes:

And the root is:

$$x = 2$$

$$x = 6$$

$$x = 12$$

$$x = 24$$

To clear $\underline{x}^2 + \underline{x}^3 = 5$ multiply by:

2

3

5

6

The 5 then becomes:

5

10

15

30

$0 \cdot x = 5$ means:

no roots

one root

every number

an error

$0 \cdot x = 0$ means:

no roots

one root

every number is a root

an error

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Linear equation	Chiziqli tenglama	Линейное уравнение
Unknown	Noma'lum	Неизвестное
Coefficient	Koeffitsiyent	Коэффициент
To transpose	Qarama-qarshi tomonga o'tkazish	Перенести
To collect	Ixchamlash	Привести подобные
To clear fractions	Maxrajdan qutulish	Избавиться от знаменателей
Standard form	Standart ko'rinish	Стандартный вид
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A linear equation can be written as:

$$ax^2 = b$$

$$ax = b$$

$$\frac{a}{x} = b$$

$$a + x = 0 \text{ only}$$

A term crossing the equals sign:

keeps its sign

changes its sign

disappears

doubles

$3x - 5 = x + 7$ gives:

$$x = 3$$

$$x = 6$$

$$x = 12$$

$$x = 1$$

To clear fractions, multiply:

the fractions only

every term

the right side only

nothing

$0 \cdot x = 7$ has:

one root

no roots

two roots

every root

The final check uses:

the last line

the original equation

a calculator

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x + 5 = 12$

ANSWER $x = 7$

2. $3x = 12$

ANSWER $x = 4$

3. $2x - 4 = 10$

ANSWER $x = 7$

4. $3x - 5 = x + 7$

ANSWER $x = 6$

5. $5x + 2 = 3x + 10$

ANSWER $x = 4$

6. $7 - x = 3$

ANSWER $x = 4$

7. $4x = 0$

ANSWER $x = 0$

Medium · the standard question

7 PROBLEMS

1. $2(x + 3) = 5x - 6$

ANSWER $x = 4$

2. $\frac{x}{2} + \frac{x}{3} = 5$

ANSWER $x = 6$

3. $\frac{x - 1}{4} = \frac{x + 1}{6}$

ANSWER $x = 5$

4. $3(x - 2) = 2(x + 1)$

ANSWER $x = 8$

5. $2(x + 3) = 2x + 5$

ANSWER No roots

6. $3(x - 1) = 3x - 3$

ANSWER Every number

7. $\frac{2x}{5} = 4$

ANSWER $x = 10$

Hard · stretch and reason

7 PROBLEMS

1. $4(x - 1) - 3(x + 2) = 0$

ANSWER $x = 10$

2. $\frac{x+1}{2} - \frac{x-1}{3} = 1$

ANSWER $x = 1$

3. $5(2x - 3) = 2(5x - 7)$

ANSWER No roots

4. $\frac{3x}{4} - \frac{x}{6} = 7$

ANSWER $x = 12$

5. $ax = 5$ with $a = 0$: how many roots?

ANSWER None

6. $0.2x + 1 = 0.5x - 0.5$

ANSWER $x = 5$

7. $2(3x - 1) - 3(2x - 1) = 1$

ANSWER Every number

07 Homework

Homework — 5 tasks

Write the check as a separate line under every answer.

1. Solve $4x - 7 = 2x + 9$.

2. Solve $3(x + 2) = 4x - 1$.

3. Solve $\frac{x}{3} + \frac{x}{4} = 7$.

4. Solve $\frac{x+2}{3} = \frac{x-1}{2}$.

5. Solve $2(x+4) = 2x+8$ and explain the answer.